ترتيب اطل على فيديو تقوية
Q1 → "Find Exam Key" سبيكتروم

58. In case of Capacitive circuit, Capacitance (C) is _____ Proportional to the Capacitive reactance (XC). 1

35

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واعدنا نبنيها

$C \propto \frac{1}{X_C}$

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سيارك ويبدع منعاً باتاً التعديل على الملف باي شكل من الاشكال ونشره من اي شخص او جهة للطلاب الا بصيغته الاصلية هذه والمنشورة من قبل فقط هذا العمل

a. Inversely proportional to inductive reactance squared
b. Directly proportional to inductive reactance
c. Inversely proportional to inductive reactance
d. Directly proportional to inductive reactance squared
e. No Effect

$X_C = \frac{1}{j\omega C}$

$C = \frac{1}{j\omega X_C}$

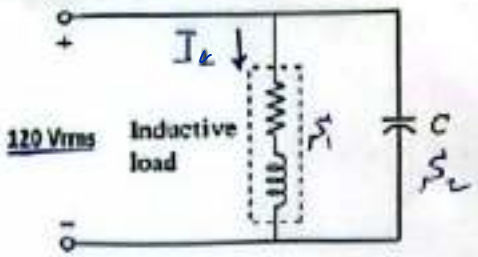
MECHATRONICS SPECTRUM TEAM

Q2

Reactive power
Q

Problem 2 [5 pts]: Consider a 60 Hz inductive load that absorbs 6000 VAR at a power factor of 0.82 lagging. If a capacitor is connected in parallel to correct the power factor to 0.93 lagging. Answer the following:

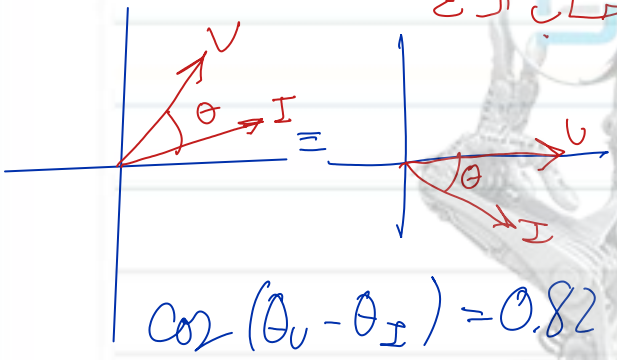
1- [2 pts] Calculate the components of the impedance of the load before connecting the capacitor bank.
2- [3 pts] Calculate the value of the Capacitance Required.



1- $Q = V_{rms} I_{rms} \text{ p.f}$

$\frac{6000}{120 \times 0.82} = I_{rms} \Rightarrow I_{rms} = 60.97$

'Z' لـ ايجاد



$\cos(\theta_v - \theta_I) = 0.82$

$-\theta_I = 34.91$

$I = \sqrt{2}\, 60.97 \angle -34.91$

$86.22 \angle -34.91$

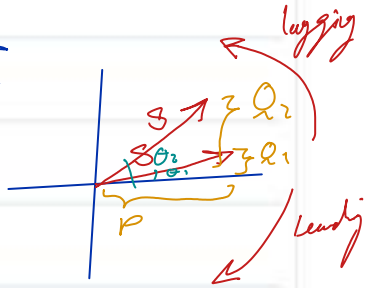
$V = 169.7 \angle 0^\circ$

$Z = \frac{V}{I} = (1.61 + j1.12)\,\Omega$

inductor

2- $C = \frac{Q_C}{\omega V_{rms}^2}$

$C = \frac{P(\tan\theta_1 - \tan\theta_2)}{\omega V_{rms}^2}$



$S = V_{rms} \cdot I_{rms}^*$

$S = \frac{V_m}{\sqrt{2}} \cdot \frac{I_m^*}{\sqrt{2}} = \frac{1}{2} V_m I_m^*$

$= \frac{1}{2}(169.7 \angle 0^\circ)(70.7 + j49.34)$

$= 5998.895 + j4185.499$

active power 'P' reative power 'Q'

$\theta_1 = 34.91 \qquad \theta_2 = 21.56$

$C = \frac{5998.895(\tan 34.91 - \tan 21.56)}{60 \times 120^2}$

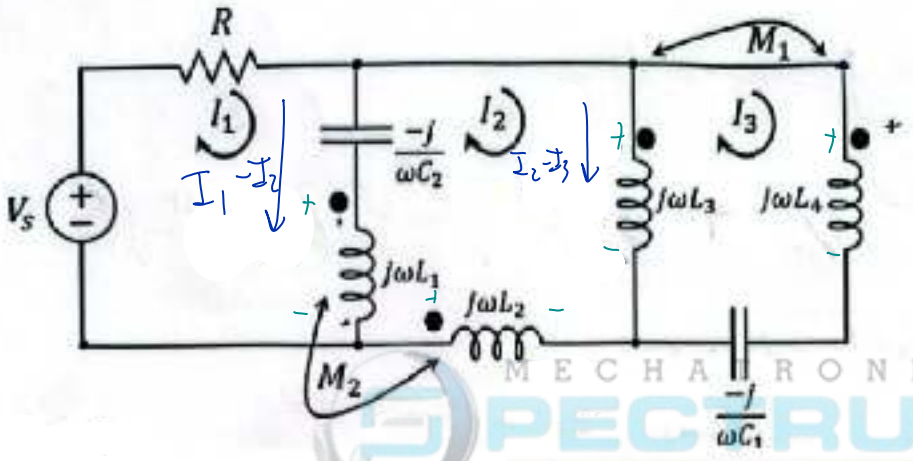
$= 2.1 \text{ mF}$

Q3

3

Problem 3 [7 pts]: Write the correct set of Mesh equations for the following circuit:

Do not solve the equations or combine any terms.



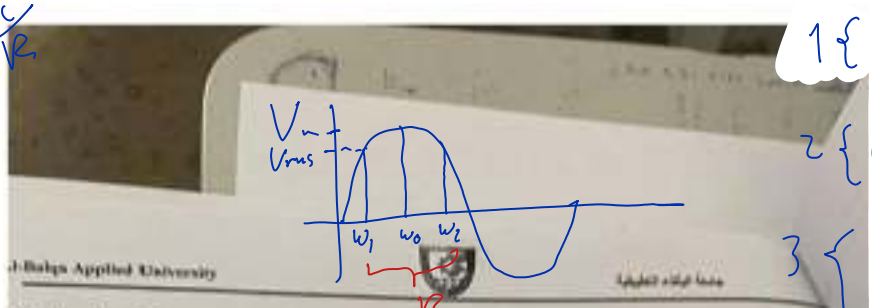
$$V_S = RI_1 + \left(\frac{j}{\omega C_2} + j\omega L_1\right)(I_1 - I_2) - j\omega M_2 I_2$$

$$j\omega L_3 (I_2 - I_3) + j\omega L_2 (I_2) + \left(j\omega L_1 - \frac{j}{\omega C_2}\right)(I_2 - I_1) = j\omega M_2 (I_1) - j\omega M_1 I_3$$

$$\left(j\omega L_4 - \frac{j}{\omega C_1}\right) I_3 + j\omega L_3 (I_3 - I_2) = j\omega M_1 I_2$$

Q4

$Q = \frac{1}{\omega_0 CR} = \frac{V_c}{R}$



Balqa Applied University

Prob.1: Consider a series RLC circuit has R =10 Ω, L = 20 mH, C = 5nF, and V_s = 5∠0° V_{rms}

1. The resonant frequency (ω_o) in (rad/sec) is:
   a. 25 K b. 100 K c. 20 K d. None of all
2. The quality factor (Q) is:
   a. 200 b. 50 c. 100 d. None of all
3. The cutoff frequencies ω_1 , and ω_2 (in rad/sec) are :
   a. (19.2, 20.8) K b. (24, 26) K c. (99.5, 100.5) K d. None of all
4. The voltage across the capacitor (V_C) at resonance (in V) is:
   a. 1000 b. 100 c. 500 d. None of all

1 { $\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(20m)(5n)}} = 100k$ rad/s

2 { $Q = \frac{\omega_0 L}{R} = \frac{100k\,(20m)}{10} = 200$

3 { $B = \frac{\omega_0}{Q} = 500$

$\omega_1 = 100k - 250 = 99.750k$

$\omega_2 = 100k + 250 = 100.25k$

4 { $V_c = I X_c = \frac{S}{R} X_c = 5 \times 200 = 1000$

$I = \frac{V_m}{\sqrt{2}R} = \frac{V_{rms}}{R}$

Q5

Prob.2: Two Loads connected in parallel across a voltage source of 100∠0° V_{rms} where the loads are:
Load 1: absorbs 250VA at 0.6 lagging p.f. and Load 2: absorbs 200W at 0.8 leading p.f

5. The current of load 1 is
   a. 1.3 − j0.75 A b. 0.75 − j1.0 A c. 1.5 − j2 A d. None of all
5. The current of load 2 is
   a. 1.3 − j0.75 A b. 0.75 − j1.0 A c. 2 + j1.5 A d. None of all
6. The complex power supplied by the source is
   a. 564.8 + j501.2 VA b. 350 + j150 VA c. 231.5 + j564.8 VA None of all
7. The power factor of the equivalent load is
   a. 0.99 lagging b. 0.748 lagging c. 0.38 leading d. None of all
8. The current supplied by the voltage source in rms is
   a. 2.822 − j2.506 A b. 2.822 A c. 3.535 A None of all
9. The impedance of the equivalent load is
   a. 24.87 − j60.67 Ω b. 28 + j4 Ω c. 60.67 + j24.87 Ω d. None of all

5. $\cos(\theta) = 0.6$ $\theta = 53.1$ $S = 250\angle 53.1$ $V_m I_m^* = 250\angle 53.1$

6. $200 = V_m I_m \text{ p.f}$ $\frac{200}{0.8 \times 100} = 2.5 = I_{rms}$ $I_{rms}^* = 1.5 + j1.99$ A $I_{rms} = 1.5 - j1.99$ A

7. $S_{total} = 250\angle 53.1 + 100 \times (2 - j1.5)$
$= 350 + j50$ VA

8. $353.55\angle 8.13$ → p.f = 0.98

9. $350 + j50 = 100\, I_{rms}$ $= 3.5 + 0.5j$ $3.5 - 0.5j = 3.535$

10. $Z = \frac{100}{3.5 - j0.5}$
$= 28 + j4\ \Omega$

Q6

Prob.3 : Consider the circuit shown in Figure 1where V = 10∠-60° V_{rms} , and I_s = 5∠10° A_{rms} answer questions 11-13.

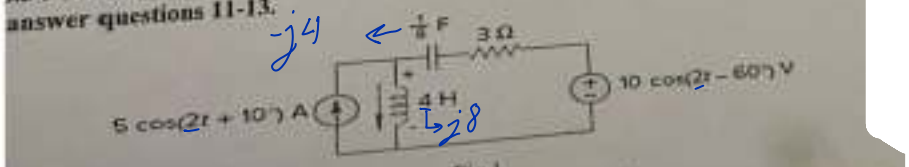


Fig.1

11. The voltage across the inductor due to V source only (in V) is:
   a. 12.9-j8.29 14.71-j6.29 c. 12+j5 d. None of all
12. The voltage across the inductor due to I_s source only (in V) is:
   a. -12.79-j6.26 b. 40 39.76-j4.36 None of all
13. The current through the inductor due to both sources in A is:
   a. 8.71+j6.29 b. 1.33-j6.81 c. 15.4-j31

11. $V = 10\angle -60° \frac{j8}{j8 - j4 + 3}$
$= 14.71 - j6.29$ V

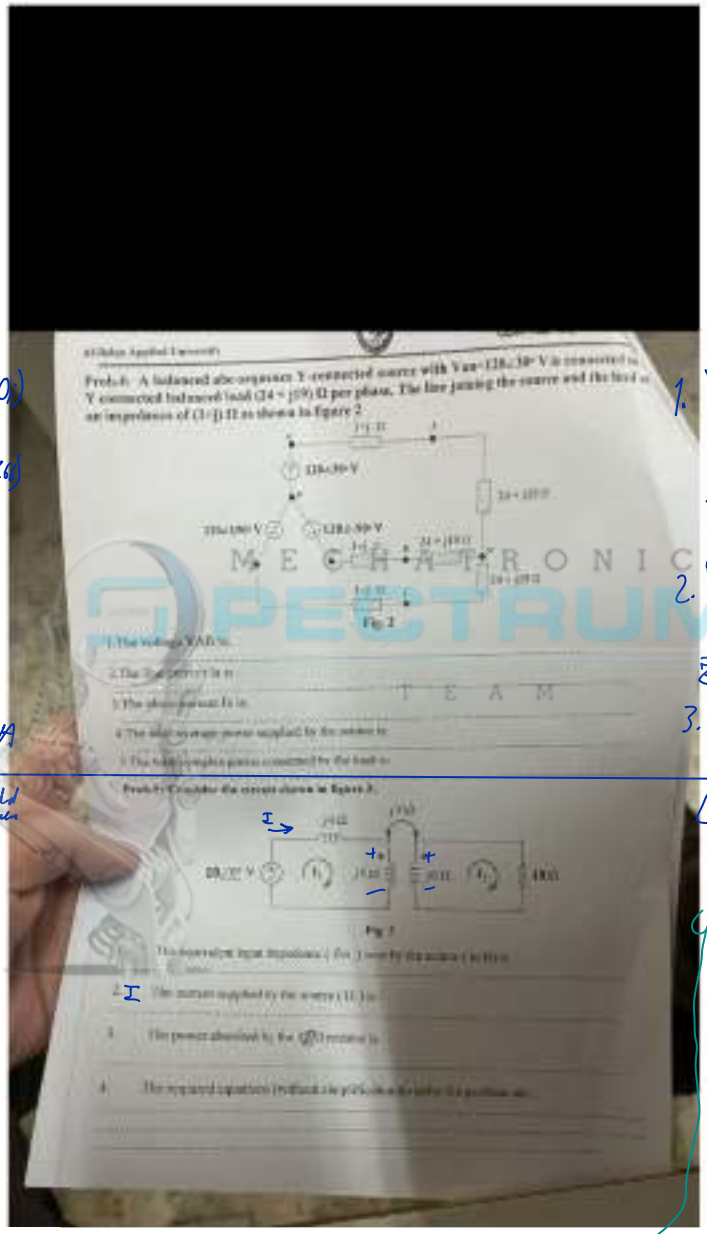
12. $I = 5\angle 10° \frac{3 - j4}{j8 - j4 + 3}$
$= -0.545 - 4.97j$
$I \times j8 = 39.76 - j4.36$

13. $I = \frac{14.71 - j6.29}{j8} = -0.785 - j1.84 + 0.545 - 4.97j$
$= -1.33 - j6.8$

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Page 2 of 2

5

Q7



4. $P = 3 \times V_{ph} \times I_x \cos(\theta_v - \theta_I)$

$\cos(30 - -8.65)$

$3 \times 120 \times 3.748 \times 0.7869$

$= 7053.7\ W$

5. $S = |I_a|^2 Z_{Ld}$

$= (7.74)^2 (24 + j19)$

$\approx 337.08 + j266\ VA$

1. $V_{AB} = V_{an} - V_{bn}$

$V_{AB} = 120\angle 30 - 120\angle -90°$

$V_{AB} = 207.84\angle 60°$

2. $V_{an} = I_a (Z_{line} + Z_{Load})$

$\frac{V_{an}}{Z_{total}} = I_a \mid I_a = 3.74\angle -8.65$

3. $I_b = I_a \angle -120$

$I_b = 3.74\angle -128.65°$

Q8

1. $Z_{in} = \frac{\text{Primary impedance} + \text{reflected impedance}}{}$

$= j9 + \frac{\omega^2 M^2}{18 + j6}$

$j3 = j\omega M$

$3 = \omega M$

$\omega^2 M^2 = (\omega M)^2$

$= j9 + \frac{3^2}{18 + j6}$

$= 8.86\angle 87.089$

2. $I = \frac{V}{Z_{in}}$

$= \frac{10}{8.86\angle 87.089} = 1.12\angle -87A$

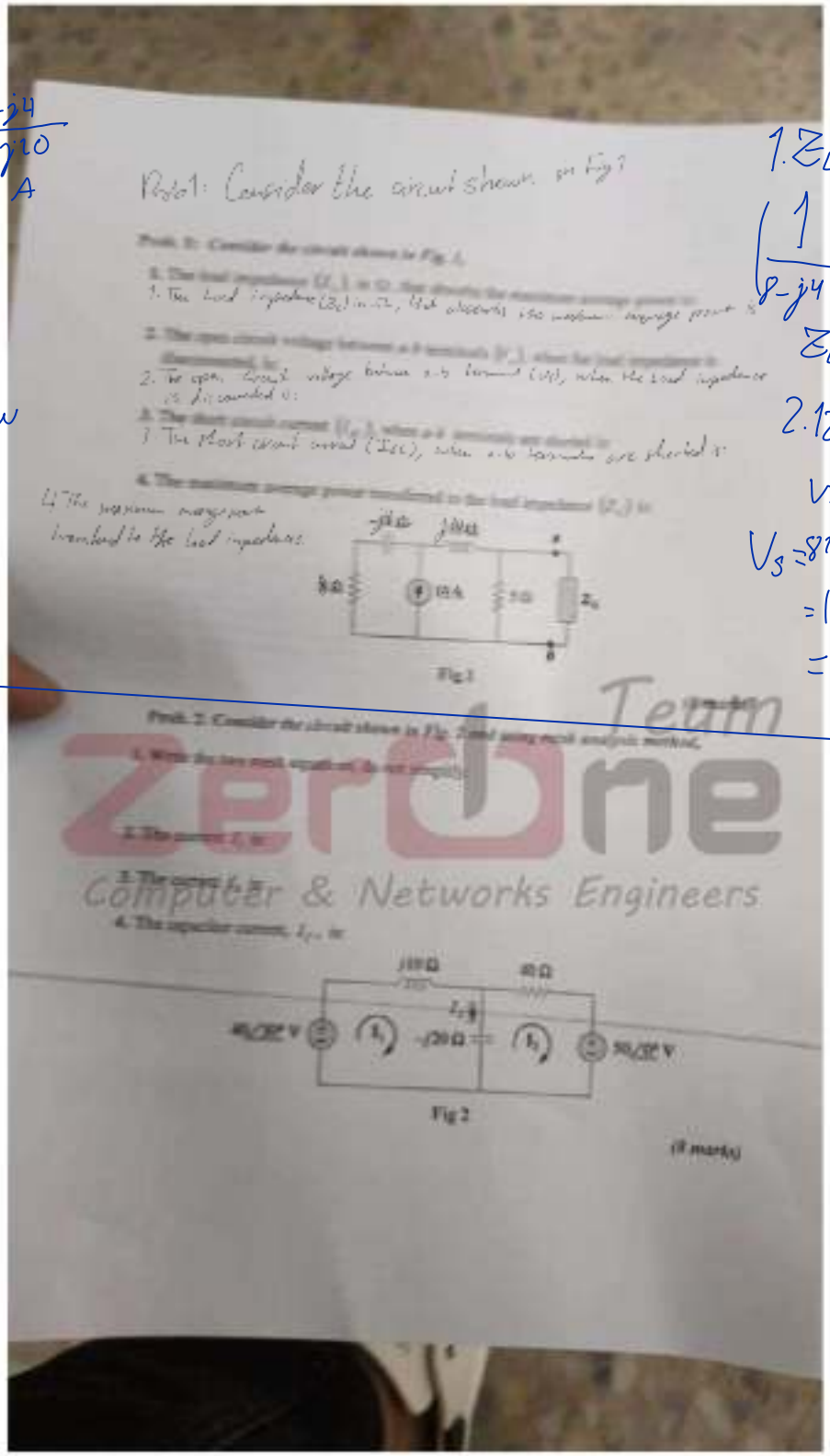
3. $P = I_2^2 R$

$I_2 = 0.17\angle -0.02$

$P = I_2 \times 18 = 0.57\ w$

4. $10 = j9 I_1 - j3 I_2$ ... ①

$18 I_2 + j6 I_2 = +j3 I_1$



3. $I_{sc} = 12\frac{8-j4}{j10}$

$= -4.8 - j9.6$ A

4. $P_L = \frac{V^2}{R}$

$= \frac{37.47^2}{6.98}$

$= 201.14$ w

1. $Z_L = Z_{th} = Z_{eq}$

$\left(\frac{1}{8-j4} + \frac{1}{5+j10}\right)^{-1}$

$Z_L = 6.98 \angle 12.09 \Omega$

2. $12 = \frac{V}{8-j4} + \frac{V}{5+j10}$

$V = 81.95 + j17.56$

$V_s = 81.95 + j17.56 \frac{5}{5+j10}$

$= (23.41 - j29.26)$ v

$= 37.47 \angle -51.33$

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Prob 3.: *Two loads are connected in parallel across a voltage source of $100\angle 0^{\circ} V_{rms}$, where load 1 absorbs 500 VAR at 0.6 lagging p.f. while load 2 absorbs 400 VA at 0.75 leading p.f*

1. The impedance of load 1 (Z_1), in Ω, is:

2. The current of load 2 (I_2), in Amps, is:

3. The complex power (S) supplied by the source is:

4. The power factor of the equivalent load is:

5. The capacitance needed to raise the pf to unity is:

6. The current supplied by the source, after the power factor is corrected, is:

(12 marks)

Prob. 4. *Consider the circuit shown in Fig. 3.*

1. The equivalent impedance seen by the source, Z_{eq}, is:

2. The current supplied by the source, I_1, is:

3. The load current, I_2, is:

4. The voltage across the primary winding, V_1, is:

5. The average power absorbed by the 2Ω resistor is:

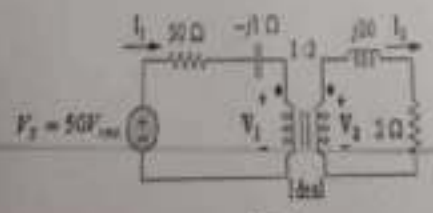
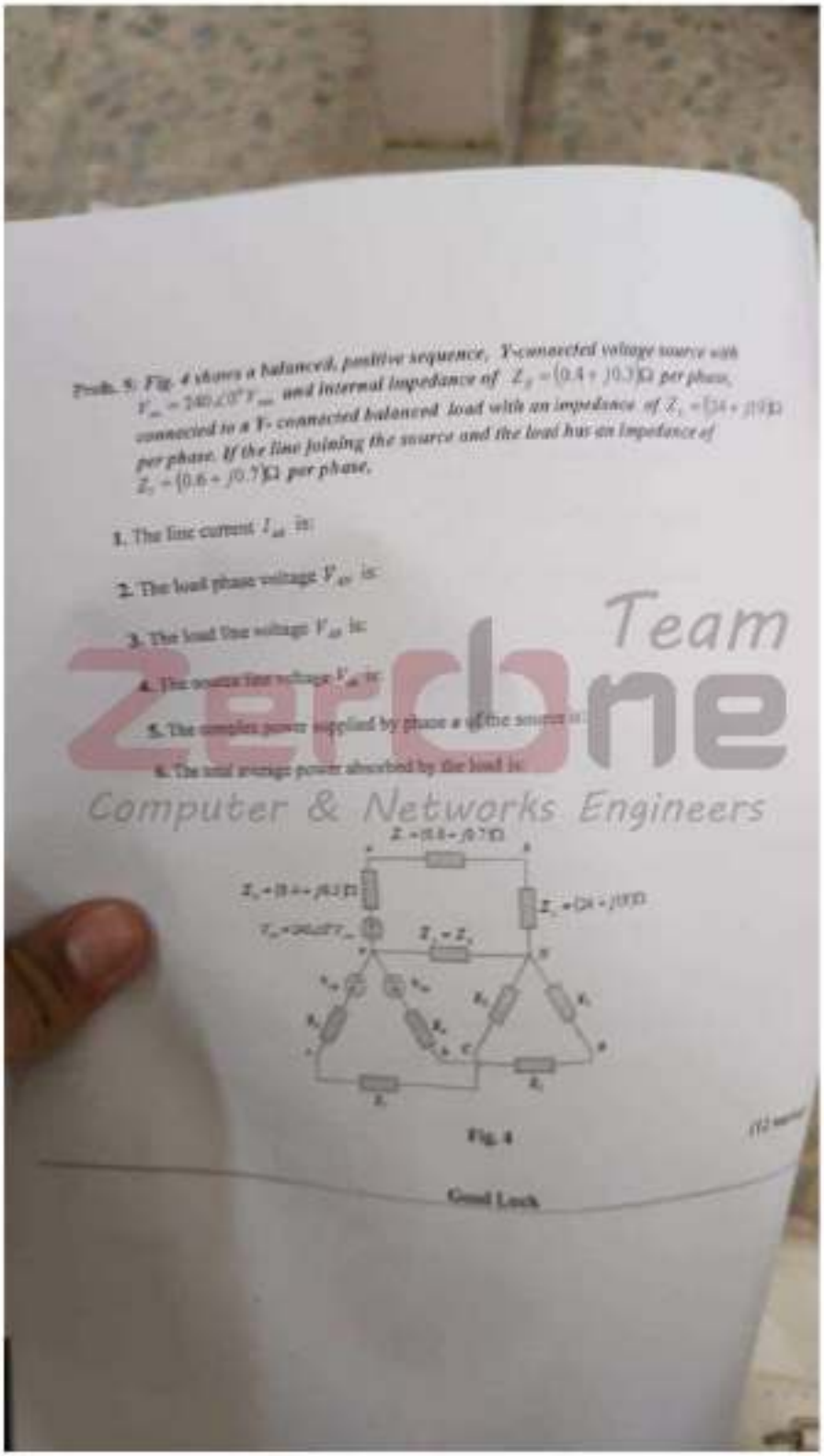


Fig.3

(10 marks)

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Prob. 5: Fig. 4 shows a balanced, positive sequence, Y-connected voltage source with $V_{an} = 240\angle 0°\,V_{rms}$ and internal impedance of $Z_g = (0.4 + j0.3)\Omega$ per phase, connected to a Y- connected balanced load with an impedance of $Z_L = (24 + j18)\Omega$ per phase. If the line joining the source and the load has an impedance of $Z_l = (0.6 + j0.7)\Omega$ per phase,

1. The line current I_{aA} is:
2. The load phase voltage V_{AN} is:
3. The load line voltage V_{AB} is:
4. The source line voltage V_{ab} is:
5. The complex power supplied by phase a of the source is:
6. The total average power absorbed by the load is:

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Fig. 4

Good Luck